

Article

Heat Exchange Analysis of Brushless Direct Current Motors

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Abstract: The brushless DC (BLDC) motor is crucial in a variety of industrial and consumer applications due to its efficiency and precise control. This study investigates the heat transfer and cooling mechanisms in liquid-cooled BLDC motors in dishwashers, which are fundamental to maintaining optimal operating temperatures. Elevated temperatures can reduce operational efficiency, emphasizing the importance of effective heat dissipation. Liquid cooling proves to be very effective and offers advantages over air cooling by providing even temperature distribution and more accurate temperature control. Integrating liquid cooling systems into dishwasher designs provides a viable solution for managing motor temperatures while preheating dishwashing water. Using existing water infrastructure, these systems dissipate heat generated during motor operation, increasing energy efficiency and reliability, as analyzed using computational fluid dynamics (CFDs). The aim of this study is to optimize thermal management strategies in BLDC motors, particularly in dishwashers, by filling a critical gap in the existing literature. The goal of this comprehensive analysis is to develop resistant and efficient cooling solutions tailored to dishwasher environments, ultimately extending the life of BLDC motors in home appliances while using heat transfer to preheat water for wash cycles.

Keywords: electric motor cooling; heat exchange; water-based cooling; stator temperature; thermal management; heat transfer; numerical methods; cooling system; computational fluid dynamics



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1. Introduction

Known for its extraordinary performance and precise control capabilities, the brushless direct current (BLDC) motor is a staple in many industrial and consumer applications. Its advantages are not only excellent performance but also energy efficiency compared to conventional motor variants [1,2]. In addition to hardware optimizations, software-level control methods play a critical role in enhancing BLDC motor efficiency and longevity. Predictive temperature control algorithms, such as those outlined in [3], enable the real-time monitoring and adjustment of motor parameters, thus improving thermal management and operational reliability. Similarly, reinforcement learning-based approaches [4] and Model Predictive Control (MPC) [5] dynamically optimize motor performance under variable load conditions. These advancements complement hardware innovations by reducing energy losses, improving torque stability and ensuring effective thermal management. This study delves into the complex domain of heat transfer and cooling mechanisms in liquid-cooled BLDC motors by presenting a comprehensive model to explain these processes.

Extensive research highlights the importance of maintaining optimal operating temperatures for electric motors, as temperature fluctuations significantly affect their performance rates [6,7]. Mechanical losses include bearing friction losses and winding losses [8]. The

operating temperature of a BLDC motor has a huge impact on its maximum power output and overall efficiency. Increased temperatures reduce the maximum power and, consequently, lead to reduced operational efficiency. This phenomenon results from the harmful effect of elevated temperatures on the insulation resistance of the windings, increasing thermal stresses and, consequently, deteriorate the overall efficiency of the motor.

The movement of heat by convection in DC motors affects the rate at which the magnetic field cavity contracts as the surface temperature increases. Understanding and effectively managing heat dissipation in liquid-cooled BLDC motors is proving to be a key effort to maintain their performance standards and extend their service life [9–12]. By comprehensively investigating the dynamics of heat transfer and cooling processes through empirical tests and CFD simulations [13,14], this study seeks to provide invaluable insights into optimizing thermal management strategies in BLDC motor applications [15–18].

The optimal operating temperatures for brushless direct current (BLDC) motors depend on design specifications and the materials used in their construction, but generally fall within a certain range to ensure their longevity and reliability. The key aspects regarding the optimal operating temperatures of BLDC motors are as follows [19]:

- Temperature range. For most BLDC motors, the optimal operating temperature is between 60 °C and 80 °C. In this respect, the motors operate efficiently and the materials used in their construction are not exposed to excessive wear or degradation.
- Maximum temperatures. Depending on the insulation class of the windings, the maximum allowable temperatures may vary. Insulation classes B (130 °C), F (155 °C) and H (180 °C) define the maximum temperatures that can be tolerated by the motor windings without permanent damage. Exceeding these temperatures can lead to insulation degradation, increasing the risk of failure.
- Critical temperatures. Temperatures above 100 °C may be critical for some motor components such as bearings, permanent magnets and electronic components. For example, neodymium magnets, commonly used in BLDC motors, may weaken their magnetic properties at temperatures above 80–100 °C, affecting motor performance.
- Ambient temperatures. The optimal operating temperature of a BLDC motor also depends on the ambient temperature. Motors designed to operate in industrial environments are often tested in ambient temperatures ranging from −20 °C to 40 °C. When operating in extreme conditions, it may be necessary to use additional cooling or heating systems.
- Cooling systems. To keep the motor in the optimal temperature range, various cooling systems are used, such as air cooling, liquid cooling and radiators. The effectiveness of these systems is crucial to ensuring stable motor operating temperatures.

In summary, optimal operating temperatures for BLDC motors are typically between 60 °C and 80 °C, with the maximum operating temperatures depending on insulation class and design. Maintaining the motor within this temperature range is crucial to its performance, reliability and durability.

The greatest heat losses occur in several key places [20,21]. The main areas where these losses are most significant are as follows.

- Stator windings:
 - Joule losses—the flow of current through the windings causes thermal losses, which are the result of the electrical resistance of the windings. These losses can be calculated using Joule’s law: $P = I^2 \times R$ (where P is the power of losses, I is the electricity and R is the resistance of the windings). High currents and resistance can lead to significant thermal losses in the windings.
- Stator core:
 - Hysteresis losses—these are caused by the cyclic magnetization and demagnetization of the stator core material. These losses are proportional to the motor operating frequency and the magnetic properties of the core material.

- Eddy currents: created by a changing magnetic field inducing currents in the conductive core material. These losses can be reduced by using steel sheets with high resistivity and thin sections, which limit the flow of eddy currents.
- Bearings:
 - Mechanical discs and friction—in BLDC motors, friction in the bearings leads to thermal losses. Although these losses are usually smaller compared to losses in the windings and core, they can become significant at high rotational speeds or with improper lubrication.
- Permanent magnets:
 - Losses in magnets—although these losses are generally small, they can occur due to eddy currents induced in the permanent magnets, especially in the case of neodymium magnets, which are relatively conductive.
- Control electronics (inverter):
 - Transistor and diode losses—the switching and conduction of semiconductor components in the inverter causes thermal losses that must be effectively dissipated using heat sinks or other cooling systems.

All these areas contribute to the total heat loss in the BLDC motor. Effective thermal management, including the appropriate selection of materials, the design of cooling systems and the optimization of motor operation, is crucial to ensuring its efficiency and durability.

Among the various methods to reduce the operating temperature of BLDC motors, liquid cooling stands out as a highly effective approach. By exploiting the inherent properties of liquids, in particular their improved thermal conductivity compared to air [22,23], liquid cooling systems enable accelerated heat dissipation from the motor to the surrounding environment. This increased heat transfer capacity not only provides more efficient cooling but also enables the BLDC motor to operate within optimal temperature ranges. The maximum permissible temperature, or rather the maximum normal heat increase, is specified in the EN 60335-1 standard, Chapter 11.8, Table 3 [24]. For the motor temperature class F (which the tested motor has), the maximum permissible temperature increase is 115 °C during operation. Based on this assumption, the initial motor temperature is 25 °C and, after adding the maximum normal temperature increase of 115 °C, the maximum allowable temperature on the motor is 140 °C. The tested BLDC motor generally showed a coil temperature of up to 140 °C; here, the iron stack is usually irrelevant as it is mainly heated by the coils, and the magnet is cooled by water.

In BLDC motors, heat transfer is a key aspect that affects their efficiency and durability. Heat transfer can be described mathematically using the equations for heat conduction and convection. The basic equation of heat conduction [25] in one dimension for homogeneous materials can be written as follows:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \quad (1)$$

where T is the temperature, t is time, x is the spatial coordinate and α is the thermal diffusion coefficient, which is defined by:

$$\alpha = \frac{k}{\rho c_p}, \quad (2)$$

where k is the thermal conductivity of the material, ρ is the density and c_p is the specific heat of constant pressure.

In the case of convection, heat transfer is described by Newton's equation:

$$q = h(T_s - T_\infty), \quad (3)$$

where q is the heat flux, h is the heat transfer coefficient, A is the heat transfer surface, T_s is the surface temperature and T_∞ is the ambient temperature.

The heat transfer coefficient h is influenced by the properties of the coolant and the flow characteristics. For forced convection in a liquid-cooled system, h can be estimated using the Nusselt number, Nu :

$$h = \frac{Nu \cdot k}{D_h}, \quad (4)$$

where Nu is the Nusselt number, a dimensionless parameter, and D_h is the hydraulic diameter of the cooling channel.

The Nusselt number for turbulent flow in a circular tube can be approximated by the Dittus–Boelter equation:

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.3}, \quad (5)$$

where Re is the Reynolds number, indicating the flow regime and Pr is the Prandtl number, a dimensionless number that represents the ratio of momentum diffusivity to thermal diffusivity.

The Reynolds number Re is calculated as:

$$Re = \frac{\rho v D_h}{\mu}, \quad (6)$$

where v is the velocity of the coolant and μ is the dynamic viscosity of the coolant.

The Prandtl number Pr is given by:

$$Pr = \frac{c_p \mu}{k}, \quad (7)$$

By combining conduction and convection, the Fourier–Biot equation can be used to describe the total heat transfer in a BLDC motor:

$$\dot{Q} = \int_A (-k \nabla T) dA + \int_A h (T_s - T_\infty) dA, \quad (8)$$

where Q is the total heat conducted through the volume.

By analyzing these equations, the cooling efficiency of a BLDC motor and an appropriate design for the cooling systems that ensure optimal motor operation under various operating conditions can be evaluated.

In BLDC motors, heat transfer involves several physical phenomena that affect the dissipation of heat generated during operation [26]. The main phenomena are discussed below:

- Heat conduction. This is the process by which heat moves through solid materials, such as the motor's core and windings. In BLDC motors, conduction mainly occurs through materials like copper in the windings and steel in the core, transferring heat from high-temperature regions to cooler areas, including the motor housing.
- Convection. Heat is transferred to the surroundings via convection. In natural convection, the warm air surrounding the motor rises and is replaced by cooler air, aiding in heat dissipation. Forced convection, using fans or cooling devices, enhances airflow around the motor, improving heat transfer efficiency.
- Thermal radiation. Heat is also dissipated through thermal radiation, where energy is emitted in the form of electromagnetic waves. Although less significant than conduction and convection in BLDC motors, thermal radiation becomes relevant at higher temperatures, especially in hot components.
- Thermal losses (Joule heating). The flow of electric current through the windings causes Joule losses, which generate heat in proportion to the square of the current and the resistance of the windings. These losses are a primary source of heat within BLDC motors.
- Magnetic losses. Hysteresis losses and eddy currents are caused by the changing magnetic fields in the motor core. These losses, which also produce heat, depend

on the magnetic properties of the core material and the frequency of the magnetic field changes.

- Heat dissipation in the motor housing. After heat is generated inside the motor (primarily through conduction), it is dissipated to the environment through the motor housing. The housing, often made of thermally conductive materials like aluminum, plays a critical role in transferring heat away from the motor to prevent overheating.

Understanding and managing these phenomena is crucial for the effective cooling of BLDC motors. Appropriate cooling system design, taking into account all of the above heat transfer mechanisms, can significantly improve motor performance and service life.

By circulating thermally conductive liquid coolant through a strategically designed system, heat generated during operation is effectively absorbed and dissipated, preventing the localized overheating and thermal degradation of critical components. Additionally, liquid cooling systems provide greater flexibility in design and implementation, allowing for the creation of solutions tailored to specific cooling requirements and environmental conditions.

Furthermore, the use of liquid cooling enables more precise control over the motor's temperature profile, facilitating dynamic adjustments to adapt to changing operating conditions and load requirements. This adaptability not only increases overall system performance but also contributes to energy efficiency and reliability by minimizing the risk of temperature-related failures or performance degradation over time.

By delving into the intricate dynamics of heat transfer and cooling processes inherent in liquid-cooled BLDC motors, this study aims to provide invaluable insights into optimizing thermal management strategies. The aim of systematic analyses and experiments is to develop innovative cooling solutions that will not only improve the performance and efficiency of BLDC motors but also use thermal transfer to preheat water. This solution can be used in household appliances such as dishwashers. The solution paves the way for progress in various application areas.

While much research has been devoted to cooling BLDC motors in automotive and electric vehicle applications, exploring cooling methods adapted to other devices (such as dishwashers) represents a promising avenue for innovation and optimization.

Dishwashers are a particularly demanding environment for BLDC motors [27], characterized by high humidity, variable temperatures and exposure to moisture and corrosive factors. These factors can significantly impact motor performance and life if not properly addressed through solid cooling strategies. Generally, tests on dishwashers are carried out using the ECO program [28].

As European regulations and consumer preferences shift toward more energy-efficient household appliances, companies must innovate to save electric energy during operations. In dishwashers, one method is to reduce heat loss and reuse heat during the washing cycle, minimizing heater operation time. This study uses CFD to enhance efficiency iteratively.

Turbulence models in computational fluid dynamics (CFD) are categorized based on their complexity and the number of additional equations. These models help to simulate the chaotic and fluctuating nature of turbulence in fluid flows. Below is an expansion of the main types of turbulence models with corresponding equations to provide a clearer understanding of their mathematical foundations [29]:

- Zero-Equation Models (e.g., Mixing Length Model)
Zero-equation models, also known as algebraic models, are the simplest form of turbulence modeling. They do not involve solving any additional transport equations and are typically used for rough estimations or in situations where computational cost needs to be minimized.

The Mixing Length Model assumes that the turbulent shear stress is proportional to the mean velocity gradient:

$$\tau_t = \rho \cdot l_m^2 \left(\frac{\partial u}{\partial y} \right)^2, \quad (9)$$

where τ_t is the turbulent shear stress, l_m is the mixing length and $\frac{\partial u}{\partial y}$ is the velocity gradient in the direction perpendicular to the flow.

This model is typically used in simpler boundary layer flows where turbulence is assumed to be primarily driven by the velocity gradient.

- One-Equation Models (e.g., Spalart–Allmaras Model)

One-equation models introduce a single additional transport equation to account for the effects of turbulence. The Spalart–Allmaras Model is an example of a one-equation model. It solves a transport equation for the turbulent viscosity ν_t , which governs the diffusion and production of turbulence. The transport equation is

$$\frac{\partial \nu_t}{\partial t} + u_j \frac{\partial \nu_t}{\partial x_j} = C_{b1} \cdot S \cdot \nu_t - C_{w1} \cdot f_w \cdot \left(\frac{\nu_t}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((v + \nu_t) \frac{\partial \nu_t}{\partial x_j} \right) \right], \quad (10)$$

where ν_t is the turbulent viscosity, S is the magnitude of the vorticity, C_{b1} and C_{w1} are model constants, d is the distance to the nearest wall and σ is a turbulent Prandtl number.

This model is often used in aerospace applications due to its balance between accuracy and computational efficiency.

- Two-Equation Models (e.g., k - ε and k - ω Models)

Two-equation models introduce two additional transport equations, typically for the turbulent kinetic energy k and another variable such as the dissipation rate ε or the specific dissipation rate ω .

- k - ε Model

The k - ε model introduces two transport equations.

■ Turbulent kinetic energy k :

$$\frac{\partial k}{\partial t} + u_j \frac{\partial k}{\partial x_j} = P_k - \varepsilon + \frac{\partial}{\partial x_j} \left[\left(v + \frac{\nu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right], \quad (11)$$

■ Dissipation rate ε :

$$\frac{\partial \varepsilon}{\partial t} + u_j \frac{\partial \varepsilon}{\partial x_j} = C_{\varepsilon 1} \frac{\varepsilon}{k} P_k - C_{\varepsilon 2} \frac{\varepsilon^2}{k} + \frac{\partial}{\partial x_j} \left[\left(v + \frac{\nu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right], \quad (12)$$

where k is the turbulent kinetic energy, ε is the dissipation rate, P_k is the production of turbulent kinetic energy, σ_k and σ_ε are turbulent Prandtl numbers and $C_{\varepsilon 1}$ and $C_{\varepsilon 2}$ are constants.

- k - ω Model

The k - ω model is similar to the k - ε model but uses the specific dissipation rate ω instead of ε :

■ Turbulent kinetic energy equation:

$$\frac{\partial k}{\partial t} + u_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(v + \partial_k \nu_t) \frac{\partial k}{\partial x_j} \right], \quad (13)$$

■ Specific dissipation rate equation:

$$\frac{\partial \omega}{\partial t} + u_j \frac{\partial \omega}{\partial x_j} = \alpha \frac{\omega}{k} P_k - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(v + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_j} \right], \quad (14)$$

where ω is specific dissipation rate, α , β and β^* are model constants, and σ_k and σ_ω are turbulent Prandtl numbers for k and ω , respectively.

- Higher-Order Models (e.g., Reynolds Stress Model, LES and DNS)
Higher-order models attempt to capture more aspects of turbulence, making them more accurate but also significantly more complex and computationally expensive. These models typically resolve the individual components of the Reynolds stress tensor or directly simulate turbulence at various scales.
 - Reynolds Stress Model (RSM)
RSM solves transport equations for each component of the Reynolds stress tensor $\overline{u'_i u'_j}$ and the turbulent dissipation rate ε :

$$\frac{\partial \overline{u'_i u'_j}}{\partial t} + U_k \frac{\partial \overline{u'_i u'_j}}{\partial x_k} = P_{ij} - \varepsilon_{ij} + \frac{\partial}{\partial x_k} \left(C_{ij}^{kl} \frac{\partial \overline{u'_i u'_j}}{\partial x_l} \right), \quad (15)$$

where $\overline{u'_i u'_j}$ represents the Reynolds stresses, P_{ij} is the production of Reynolds stresses, ε_{ij} is the dissipation of Reynolds stresses and C_{ij}^{kl} is the diffusion term.

- Large Eddy Simulation (LES)
The larger turbulent eddies are resolved directly, while the smaller eddies are modeled using a subgrid-scale model. The governing equations for LES are filtered versions of the Navier–Stokes equations.
- Direct Numerical Simulation (DNS)
DNS resolves all scales of turbulence directly, solving the Navier–Stokes equations without any turbulence modeling. DNS is extremely computationally expensive and is typically used for research purposes.

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2}, \quad (16)$$

where u_i is the velocity in the i -direction, p is the pressure and ν is the kinematic viscosity.

The washing cycle is simplified to key moments, excluding non-essential components. The workflow of simulations and numerical methods is detailed, and results are compared with real-life experiments to verify accuracy. Geometric and cycle improvements show promising results. Additionally, modeling heat flow in dishwashers is crucial to optimizing energy usage and improving design efficiency [30].

Liquid cooling systems can be integrated into dishwasher configurations, using existing water infrastructure to circulate coolant through heat exchangers in the motor housing so that heat drawn from the motor heats water for use in wash cycles [31]. By using the water supply to the dishwasher, the cooling system can effectively dissipate the heat generated during motor operation, thus maintaining optimal operating temperature and maintaining motor performance for a long time.

Moreover, liquid cooling offers the opportunity to increase the overall efficiency and reliability of the dishwasher. By more effectively regulating motor temperatures, liquid cooling systems can help optimize energy consumption and reduce operating costs while ensuring consistent and reliable performance across a variety of wash cycles and load conditions.

Selecting a material with high thermal conductivity [32,33] is crucial, but it is also important to take into account manufacturability. Therefore, the selected material should strike a balance between thermal conductivity and ease of manufacture. Selecting a material that is readily available, cost-effective and easy to shape into complex geometries can streamline the manufacturing process and reduce production costs. Additionally, the material should have excellent heat transfer properties to effectively dissipate heat from

motor components. Conducting thorough research and testing of various materials can help determine the optimal choice that meets both thermal conductivity requirements and manufacturing feasibility, ensuring the successful implementation of dishwasher cooling solutions.

By focusing on cooling and heat transfer strategies tailored specifically to dishwashers, this study seeks to fill a vital gap in the existing literature and contribute to the development of more resilient and efficient BLDC motor solutions for home appliances. The goal of this comprehensive analysis and experimentation is to determine optimal cooling and heat transfer methods.

A Model-Based Design (MBD) methodology plays a crucial role in the development of household appliances and various other fields by allowing for the creation of virtual prototypes. This approach enables engineers to simulate and optimize the performance of products before physical prototypes are made, significantly reducing development time and costs. A MBD is also widely used in the automotive, aerospace and industrial automation sectors for its efficiency in enhancing product design and functionality [34,35].

2. Materials and Methods

The heart of every efficient dishwasher is the BLDC motor (Figure 1), which silently drives the hydraulic pump that powers the cleaning process. We will focus on the main BLDC motors with 80 W of power that are generally used in dishwashers. These motors produce heat that must be managed effectively. This section details the materials and methods used to develop a water-cooling system specifically adapted to BLDC motors in dishwashers.

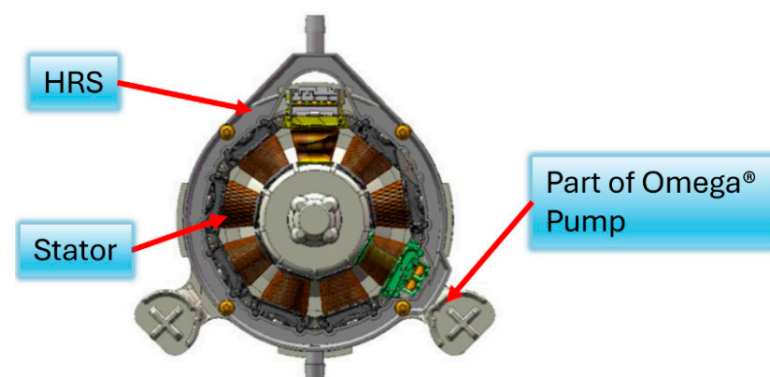


Figure 1. Three-dimensional model of a BLDC motor with a proposed heat exchanger—the Heat Recovery System (HRS).

The key to this innovation is a compact heat exchanger—a Heat Recovery System (HRS)—carefully designed to fit snugly into the stator of the motor. This is the most non-invasive and, at the same time, the closest place to the heat source in the pump. This tight integration ensures optimal heat exchange, transferring the heat generated during operation. The choice of materials for this critical element is crucial. Innovative design and flow paths in the heat exchanger further optimize heat transfer, maximizing efficiency.

However, the history of materials goes beyond the heat exchanger itself. The BLDC motor itself and the hydraulic pump are carefully selected due to the rated power, speed range and thermal properties of the motor and the type of pump used. Any surface treatments or coatings used to increase performance or durability are also carefully considered.

Combining it all together is a process of fabrication and integration. Whether it is machining, welding, brazing or even 3D printing, the methods chosen ensure the seamless integration of the heat exchanger with the motor and pump in the dishwasher system.

2.1. Materials Products and Samples

A testing methodology was developed to objectively compare the cooling and heat recovery performance of the BLDC motors under controlled conditions reflecting real-world operation. Individual systems were thermally characterized and subjected to precisely regulated environmental and load conditions in an automated test system. An overview of the process flow is shown in Figure 2.

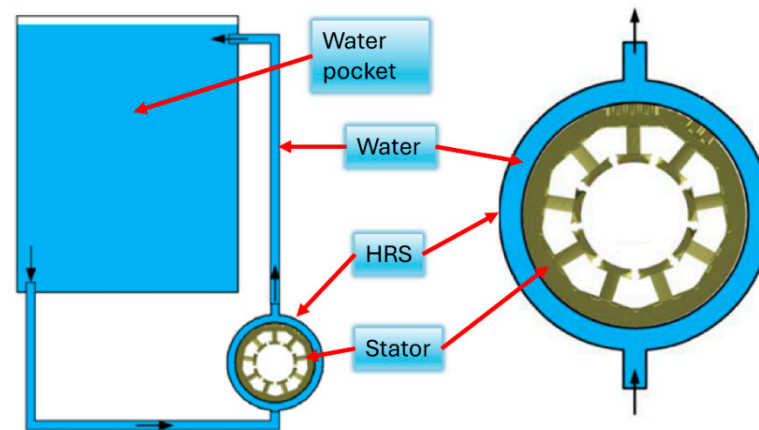


Figure 2. Test schematics representation of the Heat Recovery System (HRS).

2.1.1. Materials

Selecting the optimal material for a water-based heat exchanger system in BLDC motors involves balancing several factors:

- Thermal conductivity—This is a crucial property, since it determines the material's ability to transfer heat away from the motor. Metals generally have higher thermal conductivities than non-metals.
- Corrosion resistance—The material must be compatible with water, especially if additives or treatments are used.
- Cost—Balancing performance with affordability is essential for household appliances.
- Weight and size—Compactness is often critical in appliances.
- Easy manufacturing—Materials used in manufacturing processes should be selected with ease of manufacturing in mind. This means choosing materials that can be readily sourced, processed and manipulated with existing manufacturing technologies, minimizing production complexities, costs and time.

The choice of PET-G as the material for the Heat Recovery System (HRS) was driven by its manufacturability and suitability for rapid prototyping. PET-G, with a thermal conductivity of approximately 0.29 W/mK, was utilized to validate the HRS design in conditions reflective of real-world dishwasher operations. This decision ensured that the system could be easily tested and iteratively optimized. While higher-conductivity materials could theoretically enhance the thermal performance of the HRS, their limited availability for 3D printing and challenges in manufacturing complex geometries made PET-G the most practical option for this stage of development. This approach allowed for empirical testing under realistic constraints, aligning with the study's goal of developing a feasible and scalable solution for household appliances.

2.1.2. Sample Geometry

The Heat Recovery System (HRS) (Figure 3) is designed specifically for the tested system. The material for 3D printing used in the tests was PET-G. The dimensions of the geometry of the proposed solution were about 94 × 88 × 10 mm (without inlet and outlet—depending on the dishwasher geometry). Wall thickness was mainly 1 mm. This perfectly fits the shape adjacent to the stator of the BLDC motor. This is quite an important requirement for the system to function as efficiently as possible. A two-part manufacturing

scheme is proposed, in which the walls of the exchanger adhere to the heat source, allowing for the heat exchange.

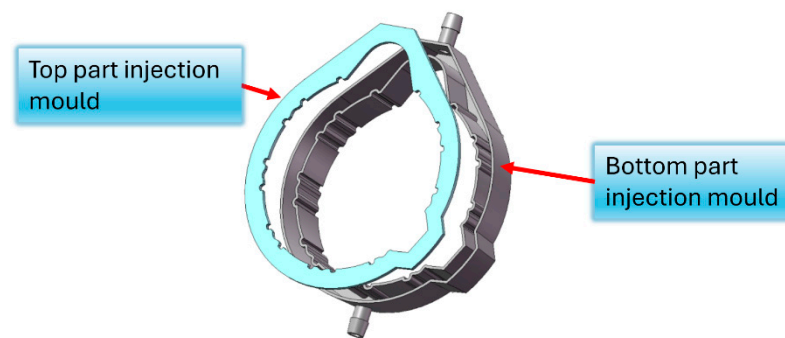


Figure 3. Three-dimensional model of the Heat Recovery System (HRS) (manufacturing version).

2.2. Equipment

Testing Equipment

The empirical experiments were undertaken using a proprietary dedicated test bench (Figure 4), where the system HRS was tested (Bleckmann GmbH, Lamprechtshausen, Austria). To meet the specific requirements of HRS testing, the tester was modified accordingly. These modifications included adapting the parameters and capabilities of the tester to analyze and evaluate the Heat Recovery System, allowing for accurate and reliable measurements. The standard Omega[®] pump test, the diagram of which is illustrated in Figure 4, is used to assess the basic operating parameters of the pumps. We use this tester to perform detailed measurements of pump performance, such as pressure and flow. Thanks to precise sensors and advanced measurement technologies, we were able to obtain accurate data that are necessary to analyze the efficiency of the pump systems. During the experiments, specialized equipment and devices were used, which are described in detail in Table 1. The tools and technologies used ensure the highest quality and precision of measurements, which is key to obtaining reliable results. Thanks to these solutions, we can fully understand the behavior of the tested systems and assess their performance and efficiency under various operating conditions. Figure 5 shows the construction and operation diagram of the test bench.

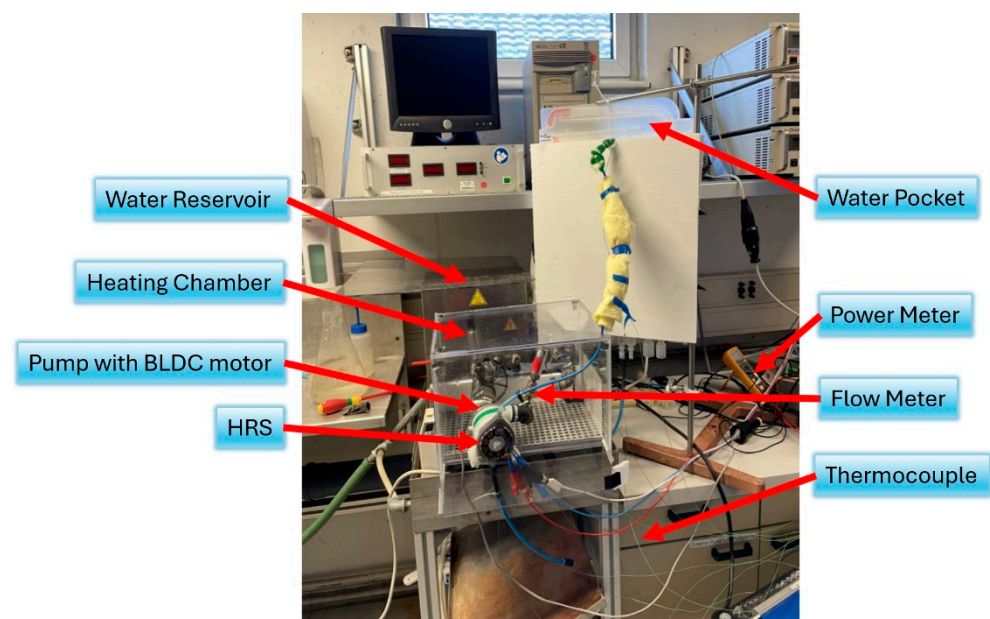
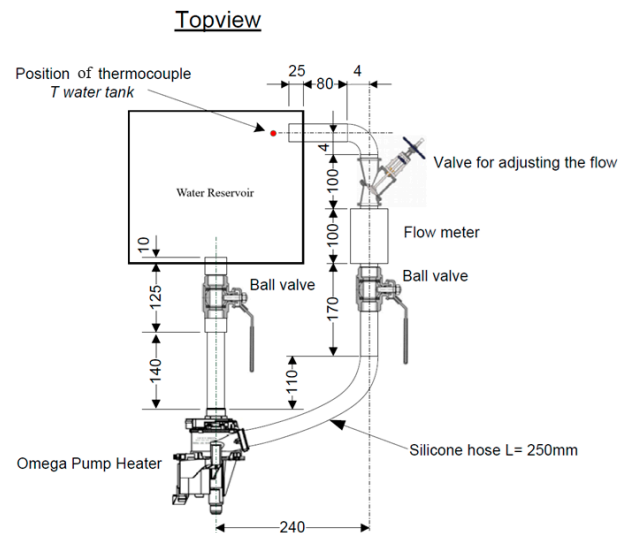


Figure 4. Photograph of the proprietary testing equipment described.

Table 1. Devices used in the test bench setup.

	Code Device	Description	Type of Device
1	TB009	Omega [®] Test Bench	Omega [®] Manual Test Bench I
2	A04043	Power Meter	N4L PPA530 ($\pm 0.1\%$)
3	A04042-1	Measurement System Rack2	NI9211 ($< 0.07\text{ }^{\circ}\text{C}$)

**Figure 5.** Diagram depicting the Omega[®] pump test bench.

The system included the following two factors:

- A test bench. We developed a test bench, replicating the operating conditions of the BLDC motor in a typical household appliance, which included:
 - A test bench with a water tank, temperature and pressure sensor and pump, on which the test was conducted (Table 1, row 1);
 - A controllable power supply to simulate motor operation at different loads and speeds (Table 1, row 2);
 - An NI module for temperature measurements, HRS and coils (Table 1, row 3);
 - Temperature sensors at key points in the motor and cooling system to measure heat transfer efficiency (Thermocouples K-NiCr/Ni class 2);
 - A flow meter to monitor the water flow rate;
 - A data acquisition system to record and analyze experimental data.
- Testing procedures. We conducted various tests under different operating conditions to evaluate the performance of the cooling system, which included:
 - Measuring steady-state temperatures at various motor working points;
 - Evaluating the impact of water flow rate on heat transfer;
 - Assessing the overall system efficiency and energy consumption.

2.3. Methods

The experimental methodology involved the installation of a customized cooling and heat transfer system carefully designed to be mounted onto a BLDC motor and placed onto a controlled test platform to simulate the operating conditions of a dishwasher. This setup involves strategically placed thermocouple temperature sensors for precise temperature monitoring.

The thermal analysis included baseline temperature measurements, systematic adjustments to operating parameters and subsequent comparative evaluations. Heat transfer characterization included dissipation rate analysis, the evaluation of heat transfer coeffi-

cients and thermal resistance control. Performance evaluation was facilitated by quantifying energy consumption, confirmed by computational fluid dynamics (CFDs) simulations performed using ANSYS software (Fluent version 2022 R2, Pressure-Based Solver) (Figure 6). Optimization strategies included design improvements and innovative cooling methods, while water preheating analysis explored the prospect of using waste heat.

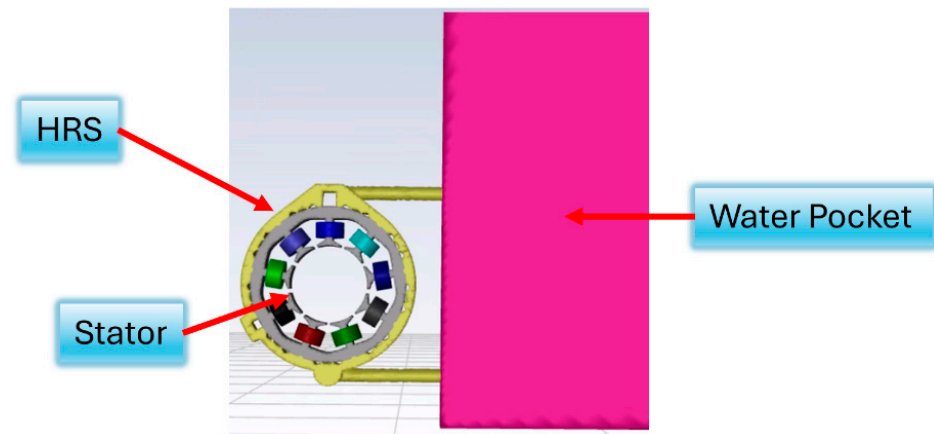


Figure 6. Three-dimensional model of the HRS used in a CFDs simulation.

The simulations conducted in this study utilized a polyhedral mesh consisting of 43,452 cells with a minimum orthogonal quality of 0.2 to ensure high accuracy, particularly in regions with significant heat exchange (Figure 7). Boundary conditions included a velocity inlet of 0.004 m/s and an outlet pressure of 0 Pa, replicating typical flow conditions in household appliances. The standard $k-\epsilon$ turbulence model was employed, with enhanced wall functions to accurately model near-wall flow and heat transfer. The thermal properties of PET-G (thermal conductivity: 0.29 W/mK) and water (thermal conductivity: 0.6 W/mK, specific heat: 4184 J/kgK) were used to represent the heat exchanger material and cooling medium, respectively.

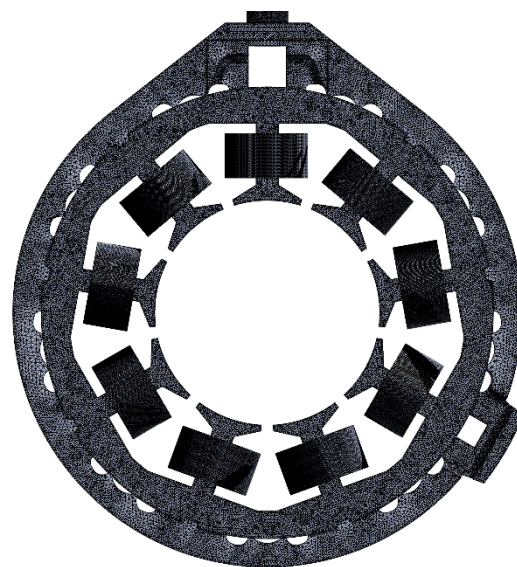


Figure 7. Finite volume mesh for HRS.

The simulation was conducted under steady-state conditions, with convergence achieved when residuals fell below 10^{-6} for energy and 10^{-4} for momentum. These parameters were chosen to replicate the operational environment of a dishwasher as accurately as possible.

These methods were developed to comprehensively investigate heat transfer and cooling mechanisms, optimize heat management strategies and increase the efficiency and reliability of dishwasher motors. The methodology diagram presented in Figure 8 comprehensively illustrates the testing process.

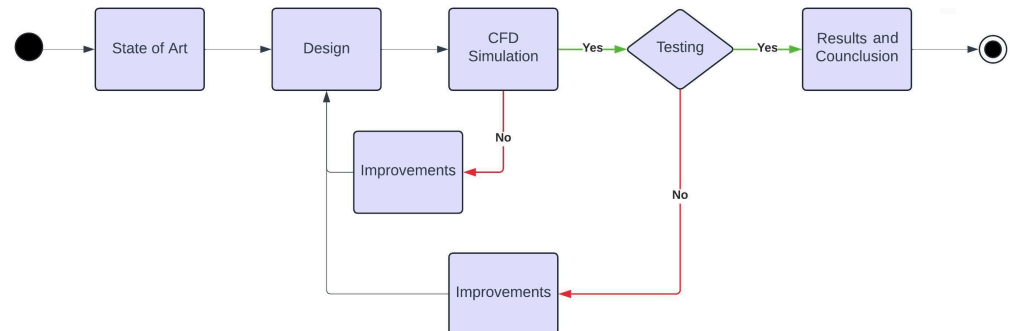


Figure 8. Flowchart depicting the methodology.

HRS Testing Cycle

The test begins by filling the water pocket with 3.1 L of water. The BLDC motor pump was started on an Omega[®] test stand. Water flowed out of the water pocket based on the principle of free convection. The motor with the HRS was placed in the heating chamber of the test stand and connected to the water tank through holes. Thermal paste was applied between the HRS and the motor stator to improve thermal conductivity.

The heating chamber is used to simulate the ambient conditions of the motor at the bottom of the dishwasher. The tested motor is VDE certified for a maximum ambient temperature of 60 °C (T60). The maximum normal heat rise in the motor needs to be tested during normal operation in the conditions where the maximum temperatures can occur. This is when the motor reached thermal steady-state condition, i.e., when the motor runs at nominal power (80 W), at a maximum ambient temperature of 60 °C and at maximum medium temperature of 75 °C. The water reservoir heated up to 55 °C imitates water heated to the temperature of the water in the dishwasher during the ECO cycle. The water in the water pocket should be at room temperature because the pocket is located near the outer wall of the dishwasher. The pump is prepared for the operating point corresponding to the rated motor power, which for the purposes of this measurement is 80 W, i.e., the input power of the motor. Thermocouples are installed on the HRS and water tank inlets, as well as on the coil and stator stack. The duration of the operation was approximately four hours.

The test was performed using the ECO parameters of the dishwasher program, because the aim of the HRS is to recover energy in the dishwasher in order to reach or contribute toward the next best Energy Efficiency Class. According to the respective standards, the Energy Efficiency Class is measured with ECO program.

Dishwasher tests are often carried out on ECO programs for several reasons:

- **Energy efficiency**—ECO programs are designed to use a minimum amount of energy, which is important for assessing the energy efficiency of the device. The results of these tests allow manufacturers and consumers to understand how much energy a dishwasher uses during a standard washing cycle.
- **Standards and Regulations**—Many regulatory tests, such as those performed to European or other international standards, require the use of ECO programs. This makes test results comparable between different dishwasher models and brands.
- **Water consumption**—ECO programs are also optimized for water consumption. Testing on these programs helps assess how effectively the dishwasher uses water, which is key to assessing its ecological and economic efficiency.
- **Typical conditions of use**—ECO programs are often chosen by users who want to save energy and water. Testing under these conditions reflects consumers' actual use of the dishwasher, resulting in more realistic and practical results.

- CO₂ emissions reduction—ECO programs are designed to minimize carbon emissions. Testing dishwashers on these programs helps assess their impact on the environment, which is increasingly important in the context of global efforts to reduce greenhouse gas emissions.

3. Results

A few shape concepts were benchmarked. The results obtained were based on empirical research with comparisons of CFD simulations. The heat exchange was between the motor and cooling liquid, i.e., tap water.

The shape of the prototype was selected for efficiency and ease of production. The CFD temperature simulation is shown in Figure 9. The flow simulation and velocity parameters are shown on Figure 10 and on Table 2. The results were optimistic, i.e., we gained 20 °C (ΔT) and 72 Wh, which are determined from:

$$Q = mc\Delta T \quad (17)$$

where m is the mass (of water 3.1 L), c is the specific heat (for water, it has the value of 4184 J/kgK) and ΔT is the temperature difference (20 °C).

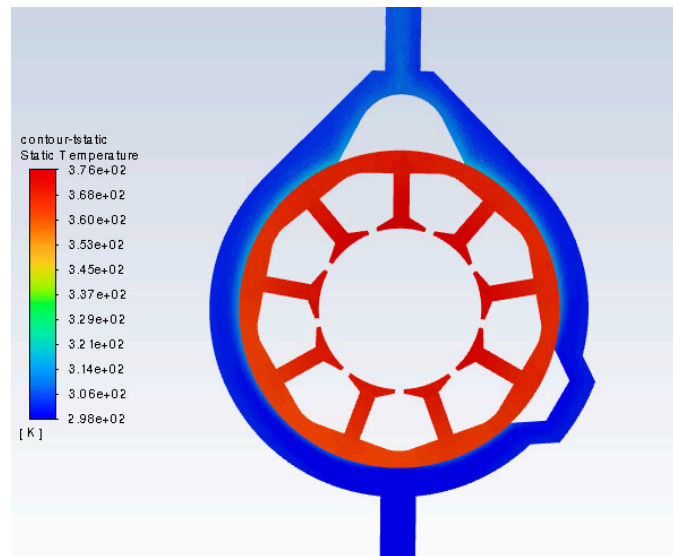


Figure 9. Temperature map on the cross-section of HRS.

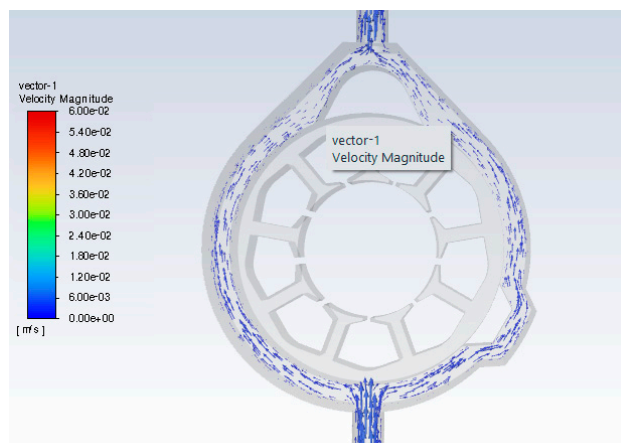
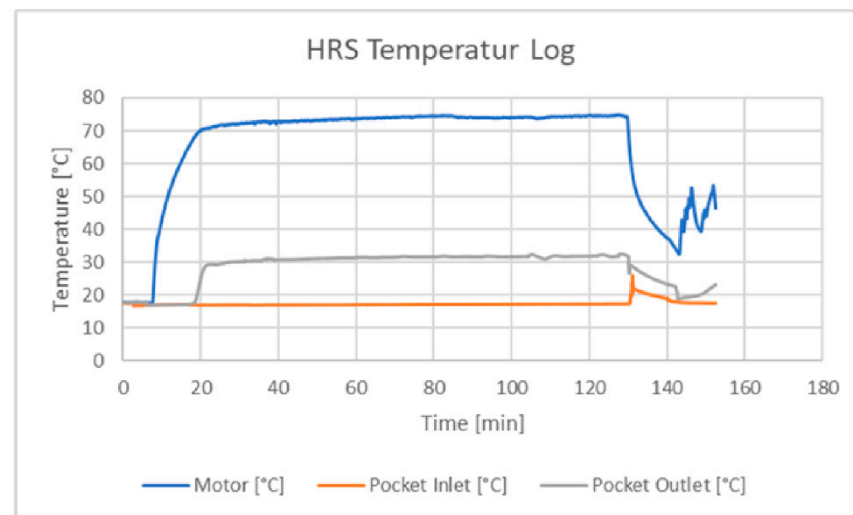


Figure 10. Map of velocity vectors on the cross-section of HRS.

Table 2. Velocity parameters in the simulation.

Mass Flow Rate	Value	Unit
Fluid inlet	0.00023391044	kg/s
Fluid outlet	−0.00023390589	kg/s
Net	4.5442482×10^{-9}	kg/s
Area-Weighted Average Velocity Magnitude	Value	Unit
Fluid inlet	0.0038433596	kg/s
Fluid outlet	0.0038363202	kg/s

The empirical verification process shows fewer results than that presented in Figure 11. During circulation, the water had an initial temperature of about 17 °C. After the heat exchange, the water reached the temperature of 30 °C. The setup parameters of the test bench are shown in Table 3.

**Figure 11.** Results of the HRS heat exchange at room temperature with cold water.**Table 3.** Parameters of test setup.

Parameter	Value	Unit
Motor Nominal Power	80	W
Water Pocket Volume	3.1	L
Initial Temperature	17 ± 0.07	°C
Final Temperature	30 ± 0.07	°C
Test Time	4	h
Energy Saving per Cycle	47 ± 0.36	Wh

A motor without cooling would have in reference to the heat rise test, a maximum coil temperature of about 140 °C and a 100 °C to 110 °C stack iron temperature. In contrast the maximum coil temperature in the experiment did not exceed 100 °C, showing significant motor cooling capabilities (Figure 12).

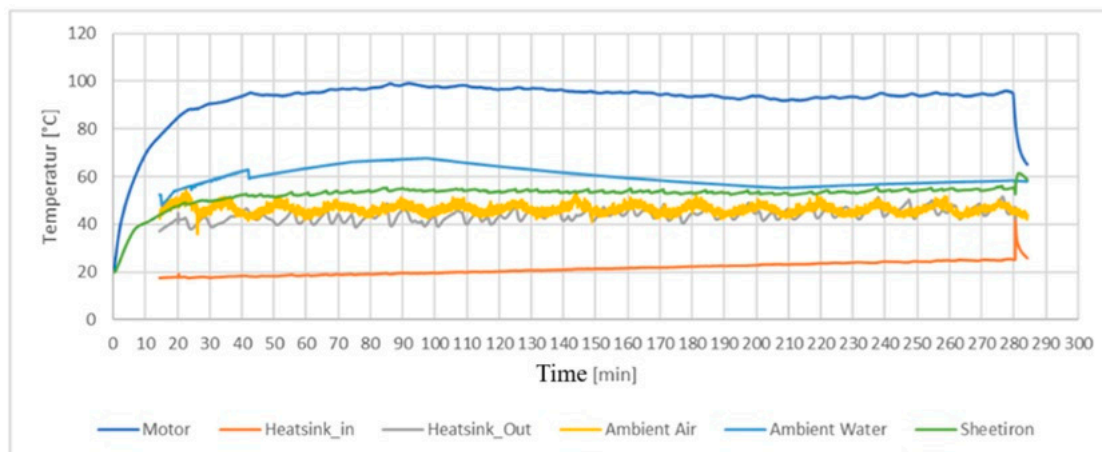


Figure 12. Results of the HRS cooling test in the heating chamber and warm water.

The cooling process has shown surprisingly good results that allow for the intended goals to be achieved.

Figure 12 demonstrates that the proposed Heat Recovery System (HRS) effectively reduces the maximum coil temperature of the BLDC motor by approximately 40 °C, lowering it from 140 °C (without cooling) to 100 °C. This reduction directly correlates to an estimated increase in motor lifespan by approximately 30%, as thermal stresses on the windings and insulation materials are significantly diminished. Furthermore, Table 3 highlights a reduction in energy consumption per cycle by 47 Wh, emphasizing the energy-saving potential of this system. Assuming an average dishwasher operates 100 cycles per month, this translates to a potential annual energy saving of 56.4 kWh, making it both an economically and environmentally sustainable solution.

The comparison between experimental and simulated results (Figures 9 and 12) further validates the effectiveness of the proposed system. While the simulations predicted a 20 °C temperature rise in the coolant, empirical data recorded a 13 °C rise under typical operating conditions. This slight deviation can be attributed to variations in real-world conditions, such as thermal paste application inconsistencies and external heat losses, which are challenging to account for in simulations.

The system's performance was also benchmarked against other cooling methods. For example, air-based cooling systems (as described in [22]) achieved lower reductions in motor temperature due to the inherent limitations of air's thermal conductivity. Similarly, hybrid air–water systems (outlined in [36]) achieved comparable results in terms of temperature reduction but lacked the compact design and energy recovery functionality of the HRS. These comparisons underline the superior adaptability and dual-purpose efficiency of the proposed system.

4. Discussion

This study investigated the effectiveness of the proposed Heat Recovery System (HRS) for cooling applications, with a particular focus on heat transfer efficiency and cost considerations. The results demonstrate the promising potential of HRS in the household appliance industry.

Our findings confirm that the HRS can be effectively employed for cooling purposes. This is corroborated by the observation that the maximum motor temperature remained below 100 °C during operation (Figure 12). Lower operating temperatures translate into several advantages:

- Improved system reliability—reducing thermal stress on components, i.e., the HRS, can contribute to enhanced system longevity and reduced maintenance requirements.

- Energy efficiency—lower operating temperatures often correlate with reduced energy consumption. This translates to potential cost savings for users and environmental benefits.
- The winding wires have a positive temperature coefficient, and resistive losses in the windings decrease at lower temperatures.
- Smaller motor (fewer materials) with the same power.

This study also highlights the potential of HRS for heat recovery. The recovered heat can be utilized for heat water, as mentioned earlier. This ability to create a closed-loop system for both cooling and water heating presents a significant cost-benefit proposition. By utilizing waste heat for water heating, the overall energy consumption of a household appliance can be significantly reduced. A lower carbon footprint of the dishwasher translates to lower operational costs for users and a more sustainable design.

This investigation underscores the importance of the thermal conductivity coefficient as a key parameter in material selection for the HRS. Materials with higher thermal conductivity facilitate more efficient heat transfer within the system. This finding provides valuable guidance for optimizing HRS design and performance.

While this study establishes the effectiveness of HRS for cooling and heat recovery, further research is warranted. Future investigations could explore the following:

- Optimizing HRS design for specific applications—tailoring the HRS to different household appliances can potentially enhancing its effectiveness.
- Investigating a wider range of materials—exploring materials with even higher thermal conductivity coefficients, which could push the boundaries of HRS performance.
- Cost-benefit analysis—a comprehensive cost-benefit analysis comparing the initial investment in the HRS with the long-term operational cost savings, which would provide valuable insights for potential users.

By addressing these future considerations, researchers can further refine the HRS and solidify its position as a viable and sustainable solution for the household appliance industry.

In the context of BLDC motor cooling, our solution, which involves water-based cooling integrated into the stator housing, presents several advantages compared to other approaches found in the literature. The focus here is to analyze and discuss how our method compares with findings in the referenced studies from the article bibliography.

In comparison to study [6], which explores a generic cooling system for BLDC motors, my solution demonstrates superior heat dissipation due to the direct integration of liquid cooling within the stator housing. The study's approach uses external cooling systems that do not benefit from direct thermal contact with the most heat-intensive components, leading to slower and less efficient heat transfer. In contrast, our integrated design ensures direct contact between the coolant and the heat source, maximizing the cooling potential. While the abovementioned study's method is easier to implement, particularly in motors not designed for liquid cooling, it does not achieve the same cooling efficiency.

Moreover, study [7] conducted on permanent magnet motors with a focus on automotive applications. The study's cooling system, which uses oil as a coolant, is optimized for long-term high-temperature applications. Although oil provides certain benefits in terms of lubrication, it lacks the high thermal conductivity of water. Our water-cooled system, while slightly more complex due to the need for corrosion-resistant materials and careful fluid management, offers faster heat removal, which is particularly beneficial in household appliances where short-term, high-intensity cooling is needed during operation cycles.

In addition, the article [19] introduces air-based cooling methods to enhance BLDC motor efficiency. Air cooling is a popular approach in certain cost-sensitive applications but does not offer the thermal performance that liquid systems like mine can achieve. As demonstrated in my CFD simulations, water cooling lowers motor temperatures more effectively than air cooling, especially under sustained loads. While air cooling has the advantage of lower complexity and cost, it struggles in applications where high power density and compact designs necessitate more aggressive thermal management.

Furthermore, study [33] focuses on thermal management in induction motors using a combined air and water cooling system. Their research shows that water cooling provides a significant reduction in motor temperature compared to air cooling alone, lowering the maximum temperature by 50% for motors with aluminum housings. This aligns with the results from our study, where water-based cooling showed superior heat dissipation. The combination of air and water cooling in study's study highlights the potential for hybrid systems; however, in our specific case, the integration of water cooling into the stator housing ensures more direct heat transfer and greater efficiency in a compact environment like household appliances.

Lastly, the work [11], which investigates CFD-based cooling designs for electric vehicles, shows that advanced computational methods can predict and enhance cooling performance in complex environments. Our approach builds on these insights by applying CFD not only for design validation but also for optimizing the shape and flow characteristics of the cooling channels within the stator housing. The study focuses on air and hybrid systems, but our approach leverages water cooling's inherent advantages in heat capacity and thermal conductivity to achieve a more efficient cooling solution.

In conclusion, while air cooling offers simplicity and oil cooling is effective in certain high-temperature applications, the water-based cooling method we developed provides superior performance in household appliances like dishwashers. This method ensures that BLDC motors operate within optimal temperature ranges, enhancing reliability and extending their service life. As seen in the comparison, our integrated solution stands out for its direct, efficient heat transfer, which is particularly crucial in environments where space and thermal management are critical.

5. Conclusions

In summary, this study elucidates the complex dynamics of heat transfer and cooling mechanisms in liquid-cooled BLDC motors, with particular emphasis on dishwasher applications. Through comprehensive analysis and experimentation, we demonstrated the effectiveness of HRS-integrated liquid cooling systems in managing motor temperatures while preheating water for wash cycles. Our findings highlight the significant advantages of using liquid cooling systems with HRS, demonstrating improved energy efficiency, reliability and overall performance of BLDC motors in demanding environments. We achieved a 40 °C reduced motor operating temperature, which provides the potential to increase motor power by 30% (from 140 °C to 100 °C).

This research highlighted the importance of selecting materials with high thermal conductivity that are easy to mass produce, further optimizing the effectiveness of cooling solutions. Utilizing existing water infrastructure, the integrated cooling and Heat Recovery System not only dissipates the heat generated during motor operation but also maximizes the use of waste heat to preheat the water, thereby increasing the overall efficiency of the dishwasher. During our tests, the heat gain was approximately 13 °C. This means that the customer can save 47 Wh hour of usage at 80 W. Assuming that, on average, a dishwasher consumes about 64 kWh per 100 cycles, the customer could save 4.7 kWh per 100 cycles.

Furthermore, this study addressed a fundamental gap in the existing literature by adapting cooling strategies and heat transfer mechanisms specifically for dishwasher environments. By providing valuable information on optimizing thermal management strategies in BLDC motors, our research contributes to the development of more resilient and efficient motor solutions for home appliances.

Going forward, continued research and innovation in this field holds great potential for further advancements in cooling technologies, ultimately leading to increased efficiency and extended service life of BLDC motors in a variety of applications. Through the collaborative efforts of academia, industry and policymakers, we can continue to drive progress in thermal management strategies, paving the way for more sustainable and efficient device designs in the future.

6. Patents

EP 3 944 473 A1 “BLDC motor with heat recovery system”.

Author Contributions: Conceptualization, M.M., W.S., C.H. and H.U.; Methodology, M.M., W.S., C.H. and H.U.; Validation, M.M. and M.K.; Formal analysis, M.M.; Investigation, M.M.; Resources, W.S.; Data curation, M.M., W.S., R.K., C.H. and H.U.; Writing—original draft preparation, M.M.; Writing—review and editing, M.M., W.S., R.K., C.H. and H.U.; Visualization, M.M.; Supervision, W.S. and R.K.; Project administration, D.K.; Funding acquisition, W.S. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: Authors Maciej Mazur, Maciej Kobielski, Damian Kadzielawa were employed by the Sanhua-Aweco. Authors Clemens Haas and Hubert Unterberger were employed by the Bleckmann GmbH & Co. KG. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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